Density, Positions, Spacings: A Three-Layer Decomposition of the Riemann Zeros, and the Operator-Side Confirmation of Form vs. Function

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Companion to *Spectral Witness Survival* and *Spectral Rigidity and Shuffled Spacing Discriminant* —

Abstract

We decompose the first 10^5 Riemann zeros into three structurally independent layers and identify the governor of each: the *average density* is the smooth Riemann–von Mangoldt term (analysis, no arithmetic); the *positions* (the fluctuation of the counting function about its mean) are owned by the primes via the explicit formula; and the *local spacings* are pure GUE (the universality class, arithmetic-blind). The separation is sharp and scale-stable. On the full 10^5 zeros with prime-powers to 3000, the prime sum reconstructs the counting fluctuation at correlation 0.84 and explained variance 0.71, beating an amplitude-matched random-frequency control by $\mathbf{122}\sigma$; extending the prime-power sum drives explained variance monotonically from 0.64 to 0.82 with a white residual (lag-1 autocorrelation 0.057), indicating the primes own the positions entirely in the limit, the residual being truncation and Gibbs ringing at the staircase steps. Independently, no arithmetic perturbation of a self-adjoint (GUE) operator moves its *spacing* statistics toward the zeros at any strength: weak arithmetic is cosmetic, strong arithmetic breaks GUE toward Poisson, and there is no selection regime. The two findings are the same statement from opposite directions: the arithmetic is exclusively positional and invisible to the spacing law. This reproduces, from the operator/explicit-formula side and at the same 10^5 scale, the *form vs. function* categorisation obtained empirically in *Spectral Rigidity and Shuffled Spacing Discriminant* (spacing/counting/shape are form; the arithmetic witness is function), and it sharpens the Hilbert–Pólya target accordingly.

1 The decomposition

Write the exact zero-counting function as $N(T) = \langle N(T)\rangle + S(T)$, where the smooth mean is the Riemann–von Mangoldt term

$$\langle N(T)\rangle = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8},$$

and $S(T)$ is the fluctuation. The local level-spacing law is a separate object, obtained after unfolding by $\langle N\rangle$. These three — mean density, fluctuation $S(T)$, and spacing law — are structurally independent, and we find each has a different governor.

layer	quantity	governor
density	$\langle N(T)\rangle$	smooth analysis (no arithmetic)
positions	$S(T)$ fluctuation	**the primes** (explicit formula)
spacings	unfolded gap law	**pure GUE** (universality class)

The two load-bearing claims are that the positions are prime-governed and the spacings are not. We establish both, and then note they are the same statement.

2 Positions are owned by the primes (T1)

The Riemann–von Mangoldt explicit formula gives the oscillatory part of the count as a sum over prime powers,

$$S(T) \;\sim\; -\frac{1}{\pi}\sum_{p}\sum_{k\geq 1}\frac{1}{k}\,p^{-k/2}\sin\bigl(kT\log p\bigr)+\cdots$$

We evaluate the true fluctuation $S_{\text{true}}(T) = N(T) - \langle N(T)\rangle$ on a dense grid and compare it to this prime sum.

Scaling on real data. The reconstruction is flat in quality across a $50\times$ range in the number of zeros, while the significance against control grows with scale:

zeros M	primes $< P$	corr	expl. var	vs. random
2,000	300	0.840	0.705	$+54\sigma$
10,000	1000	0.848	0.719	$+106\sigma$
50,000	2000	0.838	0.702	$+95\sigma$
100,000	3000	0.840	0.706	$\mathbf{+122\sigma}$

The control is decisive. Replacing the log-prime frequencies $\log p$ with amplitude-matched random incommensurable frequencies gives correlation 0.004 ± 0.019 — nothing. It is not that any oscillatory sum fits $S(T)$; it is specifically the prime frequencies, at 122σ over the matched null at full scale. The arithmetic does real, large selection work on the positions.

Completeness (T1′). Extending the prime-power sum (more primes, higher k) drives the explained variance monotonically upward with no plateau:

$$0.643 \to 0.697 \to 0.740 \to 0.776 \to 0.801 \to 0.822$$

(primes to $3\times 10^2, 10^3, 3\times 10^3, 10^4, 3\times 10^4, 10^5$). The residual has lag-1 autocorrelation 0.057 — white. An unstructured, monotonically shrinking residual is the signature of a truncated convergent sum (with Gibbs ringing at the step discontinuities of the staircase), not a structural ceiling. We therefore read the limit as full reconstruction: *the primes own the positions entirely*, with no non-prime residual to chase.

3 Spacings are pure GUE, arithmetic-blind (T2)

We test whether arithmetic, imposed on a self-adjoint operator, can move its *spacing* statistics toward the zeros. Take a GUE core $H_0 = (A + A^\dagger)/2$ and add a prime-log diagonal at strength ε, comparing the spacing-histogram distance to the true zeros against the noise band of pure-GUE draws.

ε	distance to zeros	verdict
0 (pure GUE)	0.0455	baseline (in band)
0.05–0.2	0.044–0.047	same as GUE (cosmetic)
0.4	0.073 ($+2.1\sigma$)	breaks toward Poisson
0.8	0.312 ($+18.7\sigma$)	breaks

There is **no selection regime**. Weak arithmetic leaves the spacing law indistinguishable from generic GUE; strong arithmetic destroys GUE toward clumping. The zeros' spacing statistics are reproduced by *any* self-adjoint operator (GUE is generic) and carry no detectable arithmetic content. Two further controls corroborate the picture: (i) symmetrising a duality does not build the spacing law — four natural finite encodings of the zeros–primes coupling all miss the GUE band, and balanced involutions imposed on a GUE core push it toward Poisson, so an externally imposed self-duality *breaks* repulsion unless the involution already commutes with H; (ii) self-adjointness alone already yields the GUE band. The arithmetic is not in the spacings.

4 The two findings are one statement

Positions are prime-governed; spacings are arithmetic-blind GUE. Equivalently: *the arithmetic content of the zeros lives entirely in where the levels sit, not in how they are spaced.* This is exactly the **form/function** split of *Spectral Rigidity and Shuffled Spacing Discriminant*, now derived from the operator/explicit-formula side at the same 10^5 scale:

	empirical (*Spectral Rigidity and Shuffled Spacing Discriminant*)	operator side (this note)
form	spacing, counting, shape: 0.4–1.0σ	spacing law: pure GUE, arithmeti
function	arithmetic witness: $\sim$11,500σ	positions: prime-governed, 122σ

Two independent routes — the marginal-matched surrogate battery and the explicit-formula position reconstruction — land on identical structure: the spacing distribution is the universality class (form, shared by every GUE sequence), and the primes are the object-specific content (function), located in the positions. The agreement across methods at full scale is the principal evidence that the form/function cut is structural rather than an artifact of either method.

5 Consequence for the Hilbert–Pólya operator

The decomposition characterises H tightly, turning a string of negative results into a positive specification:

Observation 1. *If a self-adjoint H has the Riemann zeros as spectrum, then:*

(i) Spacings come for free. *Self-adjointness alone yields GUE level statistics; the spacing law imposes no arithmetic constraint on H.*

(ii) Arithmetic enters through positions. *The eigenvalue positions (the counting fluctuation) are governed by the prime sum, completely (white residual). Any construction of H must reproduce the prime-driven $S(T)$, not a spacing statistic.*

(iii) Self-duality is generated, not imposed. *An externally imposed balanced involution destroys GUE repulsion; only an involution that already commutes with H preserves it. The self-dual structure is downstream of H, so H cannot be obtained by symmetrising a duality brought to it.*

This selects the Berry–Keating–type position/counting route — an operator whose eigenvalue locations carry the prime staircase while GUE owns the local spacings — as the one consistent with every test, and rules out the symmetrisation and diagonal-perturbation routes. It does not construct H; it specifies which layer the arithmetic must inhabit (positions) and which constructions are dead ends.

6 Scope

The position reconstruction verifies the *known* Riemann–von Mangoldt explicit formula at high significance and full scale; that part is verification, not a new theorem. What is established here as a result is the *three-layer separation* and its consequence: that the arithmetic is exclusively positional and complete (white residual, no non-prime structure), that the spacing law admits no arithmetic selection at any perturbation strength, and that these two facts reproduce the independently-obtained form/function categorisation. The explained-variance limit of 1 is inferred from the monotone climb and the white residual, not from summation to infinity; the reported figure at primes $< 10^5$ is 0.82. All computations use the real first 10^5 Riemann zeros and the canonical seed 20260423.

Accompanies *Spectral Witness Survival* and *Spectral Rigidity and Shuffled Spacing Discriminant.*